Name
Date
Course

Review on Linear Equations

The Slope of a Line

If (X_1, Y_1) and (X_2, Y_2) are two points on a line, the slope of the line is defined by the equation below.

$$m = \frac{Y_2 - Y_1}{X_2 - X_1}$$

Slope measures the steepness of a line. If the slope of a line is positive, the line travels up moving from left to right. If the slope of a line is negative, the line travels down moving left to right. If the slope of a line is zero, the line is **horizontal**. If the slope of a line is undefined, the line is vertical.

The Three Forms of a Linear Equation

Standard Form	Slope-Intercept Form	Point-Slope Form
$AX + BY = C$	$Y = mx + b$	$Y - Y_p = m(x - X_p)$
A, B, and C are constants and X and Y are Variables.	m = Slope, b = y-intercept, and x and y are variables.	m = slope, X_p is the X-coordinate of a Known point, and Y_p is the y-coordinate of the same Known point. X and Y are variables.

Identify the form of each linear equation below by writing "S" for standard, "SI" for slope-intercept, or "PS" for point-slope.

① $2x + 4y = 3$ $\boxed{S}$ ② $y = \frac{5}{6}x + 3$ $\boxed{SI}$ ③ $\frac{1}{2} = \frac{8}{9}x + \frac{3}{2}y$ $\boxed{S}$

④ $y - 4 = -3(x - 5)$ $\boxed{PS}$ ⑤ $6.2x + y = -5$ $\boxed{S}$ ⑥ $y = 2x + \frac{4}{3}$ $\boxed{SI}$

Identify the slope and y-intercept of each of the following linear equations.

$$y = mx + b$$

⑦ $y = \frac{4}{3}x + 1$ $\boxed{m = \frac{4}{3}, b = 1}$ ⑧ $y = 2x + \frac{3}{8}$ $\boxed{m = 2, b = \frac{3}{8}}$ ⑨ $y = \frac{x}{2} - 4$ $\boxed{m = \frac{1}{2}, b = -4}$

⑩ $-4x - 8 = y$ $\boxed{m = -4, b = -8}$ ⑪ $y = \frac{4}{9}x$ $\boxed{m = \frac{4}{9}, b = 0}$ ⑫ $-\frac{2}{7}x + \frac{5}{6} = y$ $\boxed{m = -\frac{2}{7}, b = \frac{5}{6}}$

Write linear equations with the following slopes and y-intercepts.

$$y = mx + b$$

⑬ $m = 2, b = \frac{1}{3}$ $\boxed{y = 2x + \frac{1}{3}}$ ⑭ $m = -\frac{2}{5}, b = 4$ $\boxed{y = -\frac{2}{5}x + 4}$ ⑮ $m = -7, b = -3$ $\boxed{y = -7x - 3}$

(16) $m=\frac{3}{4}$, $b=-\frac{8}{9}$ (17) $m=\frac{1}{6}$, $b=2$ (18) $m=5$, $b=-3.7$

$\boxed{y=\frac{3}{4}x-\frac{8}{9}}$ $\boxed{y=\frac{1}{6}x+2}$ $\boxed{y=5x-3.7}$

Find the slope of each of the following linear equations and a point on the line produced by graphing the equation.

$$y-y_p=m(x-x_p)$$

(19) $y-4=\frac{2}{7}(x-5)$ (20) $y-7=-\frac{1}{3}(x+9)$ (21) $5(x-2)=y+3$

$\boxed{m=\frac{2}{7}, (5,4)}$ $\boxed{m=-\frac{1}{3}, (-9,7)}$ $\boxed{m=5, (2,-3)}$

(22) $\frac{5}{12}(x+4)=y+3$ $\boxed{m=5/12, (-4,-3)}$

In each problem below, write an equation of the line that contains the given point and has the given slope.

$$y-y_p=m(x-x_p)$$

(23) $(6,8)$, $m=4$ (24) $(-5,3)$, $m=\frac{6}{11}$ (25) $(2,-3)$ $m=-6$

$\boxed{y-8=4(x-6)}$ $\boxed{y-3=\frac{6}{11}(x+5)}$ $\boxed{y+3=-6(x-2)}$

(26) $(-1,-3)$, $m=\frac{1}{6}$ (27) $(2,-9)$, $m=-\frac{3}{5}$ (28) $(7,7)$, $m=-1$

$\boxed{y+3=\frac{1}{6}(x+1)}$ $\boxed{y+9=-\frac{3}{5}(x-2)}$ $\boxed{y-7=-(x-7)}$

$Ax+By=c$ $\quad y=mx+b$

A) Convert the following linear equations from standard form to slope-intercept form.
B) Then find the slope and y-intercept of the lines produced by graphing the equations.

(29) $4x+2y=8$

A) $2y=8-4x$

$y=4-2x$

$\boxed{y=-2x+4}$

B) $\boxed{m=-2, b=4}$

(30) $3=7x+4y$

A) $3-7x=4y$

$\frac{3}{4}-\frac{7}{4}x=y$

$\boxed{y=-\frac{7}{4}x+\frac{3}{4}}$

B) $\boxed{m=-\frac{7}{4}, b=\frac{3}{4}}$

(31) $20x-5y=25$

A) $-5y=25-20x$

$y=-5+4x$

$\boxed{y=4x-5}$

B) $\boxed{m=4, b=-5}$

(32) $-3x+2y=8$

A) $2y=8+3x$

$y=4+\frac{3}{2}x$

$\boxed{y=\frac{3}{2}x+4}$

B) $\boxed{m=\frac{3}{2}, b=4}$

A) In each problem below, use the slope equation to find the slope of the line that contains the two given points.
B) Then write the point-slope form of the equation of the line containing the two given points.
C) Then convert the equations from point-slope form to slope-intercept form.

33) $(4,7)$ & $(3,5)$
(x_1,y_1) (x_2,y_2)

A) $m = \frac{y_2-y_1}{x_2-x_1} = \frac{5-7}{3-4} = \frac{-2}{-1}$
$= \boxed{2}$

B) $y - y_p = m(x - x_p)$
$\boxed{y-7 = 2(x-4)}$

C) $y-7 = 2(x-4)$
$y-7 = 2x-8$
$\boxed{y = 2x-1}$

34) $(6,-2)$ & $(1,8)$

A) $m = \frac{8-(-2)}{1-6} = \frac{10}{-5}$
$= \boxed{-2}$

B) $\boxed{y+2 = -2(x-6)}$

C) $y+2 = -2(x-6)$
$y+2 = -2x+12$
$\boxed{y = -2x+10}$

35) $(7,11)$ & $(3,-5)$

A) $m = \frac{-5-11}{3-7} = \frac{-16}{-4}$
$= \boxed{4}$

B) $\boxed{y-11 = 4(x-7)}$

C) $y-11 = 4(x-7)$
$y-11 = 4x-28$
$\boxed{y = 4x-17}$

36) $(4,1)$ & $(-2,-3)$

A) $m = \frac{-3-1}{-2-4} = \frac{-4}{-6}$
$= \frac{2}{3}$

B) $\boxed{y-1 = \frac{2}{3}(x-4)}$

C) $y-1 = \frac{2}{3}(x-4)$
$y-1 = \frac{2}{3}x - \frac{8}{3}$
$y = \frac{2}{3}x - \frac{8}{3} + 1$
$\boxed{y = \frac{2}{3}x - \frac{5}{3}}$

$\frac{2}{3}(4)$
$\frac{2 \to 4}{3 \to 1} = \frac{8}{3}$

$-\frac{8}{3} + \frac{1}{1}$
$-\frac{8}{3} + \frac{3}{3}$

Review on Graphing Linear Equations

Four Methods

I) Finding two random points.

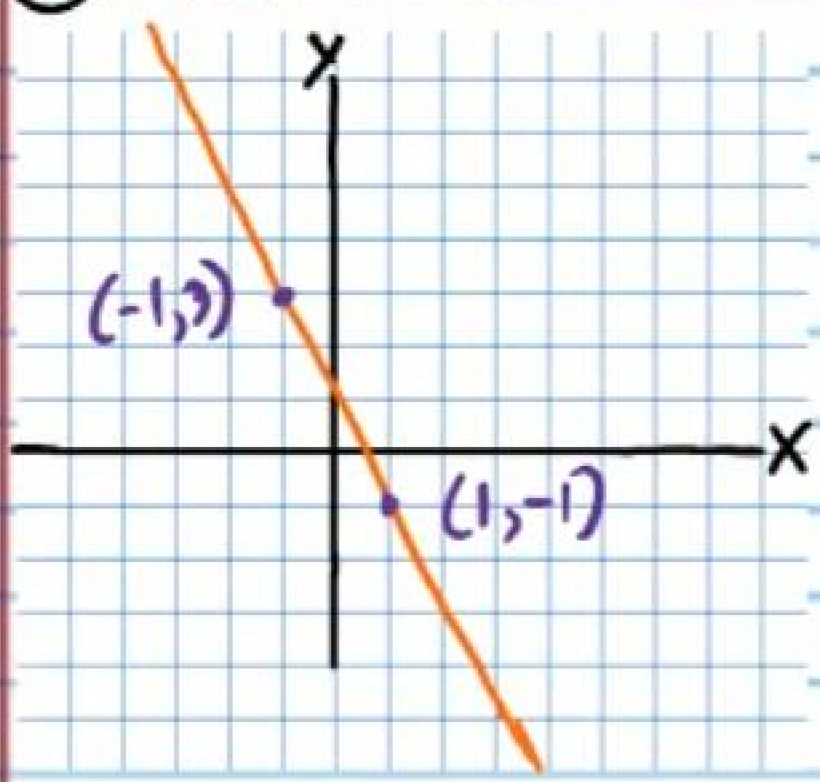
II) Finding X and y intercepts.

III) Converting to slope-intercept form and using slope and y-intercept.

IV) Using point-slope form.

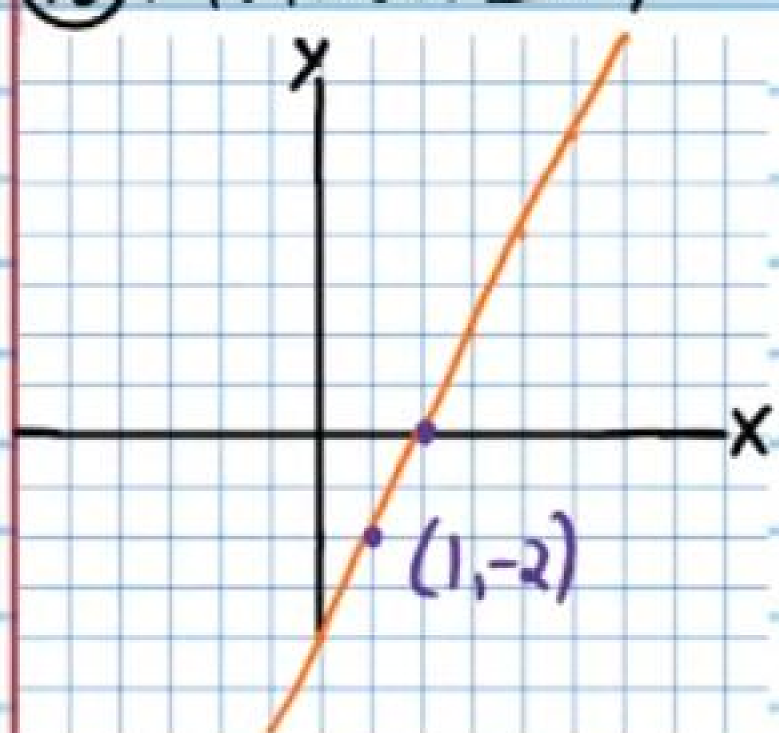
Graph the following equations using the indicated method.

(37) Method I: $2x+y=1$



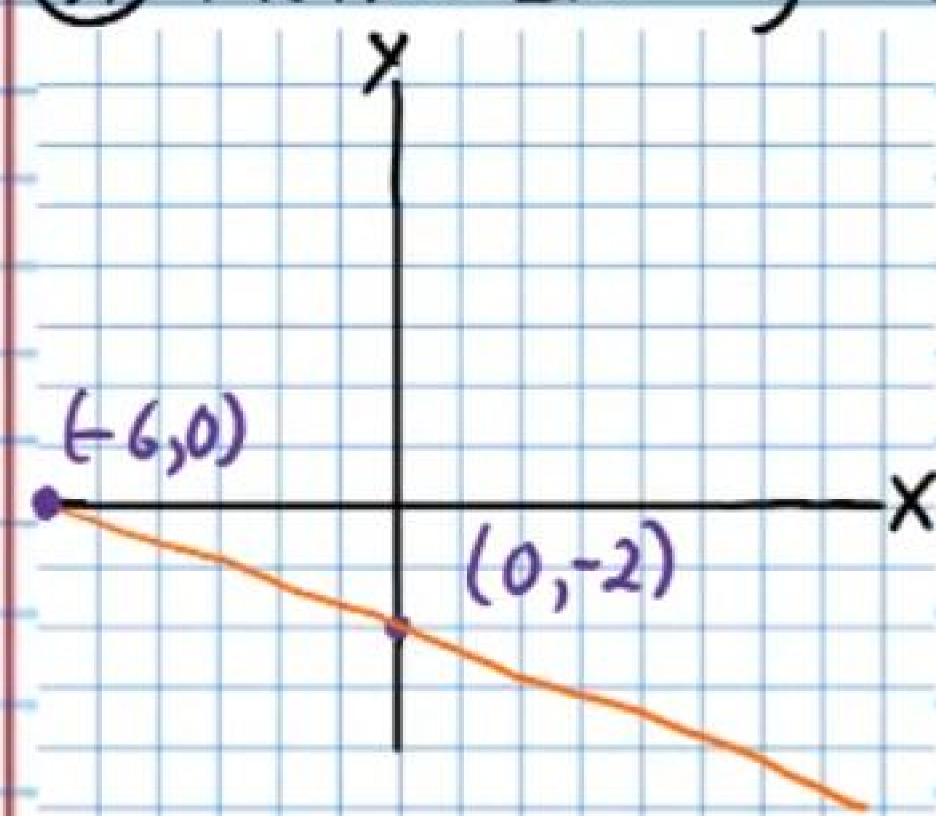
Point I	Point II
Choose $x=1$	Choose $x=-1$
$2(1)+y=1$	$2(-1)+y=1$
$2+y=1$	$-2+y=1$
$y=-1$	$y=3$
$(1,-1)$	$(-1,3)$

(38) Method I: $y=2x-4$



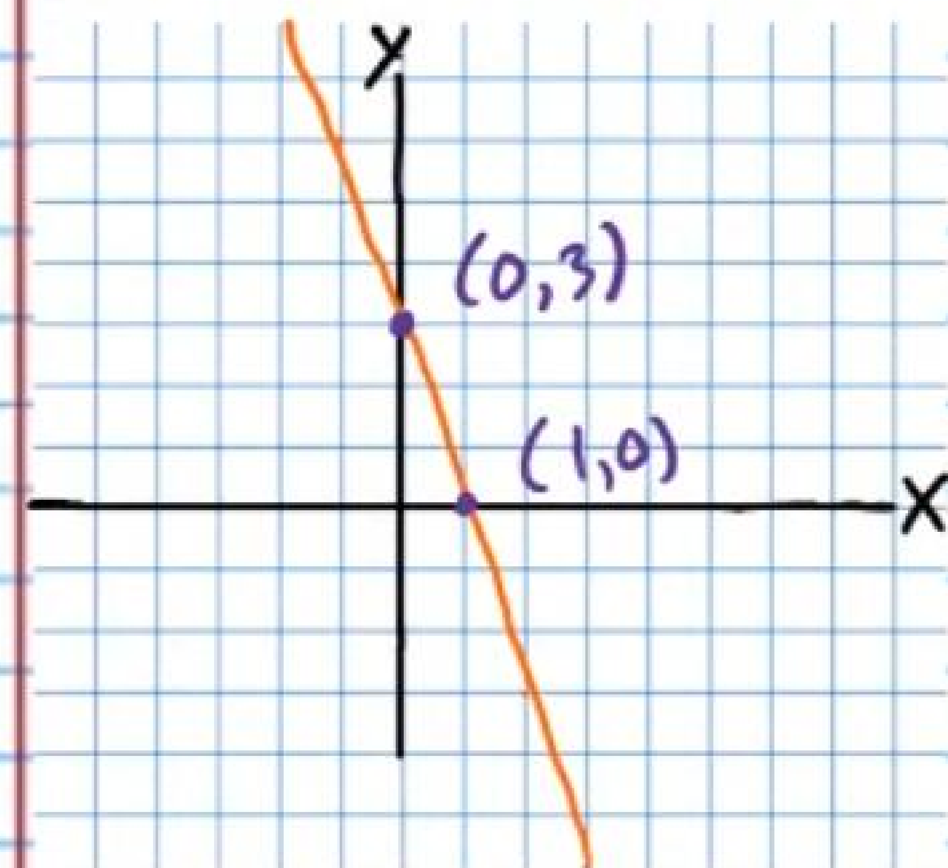
Point I	Point II
Choose $x=1$	Choose $x=2$
$y=2(1)-4$	$y=2(2)-4$
$y=2-4$	$y=0$
$y=-2$	
$(1,-2)$	$(2,0)$

(39) Method II: $-\frac{x}{3}-2=y$



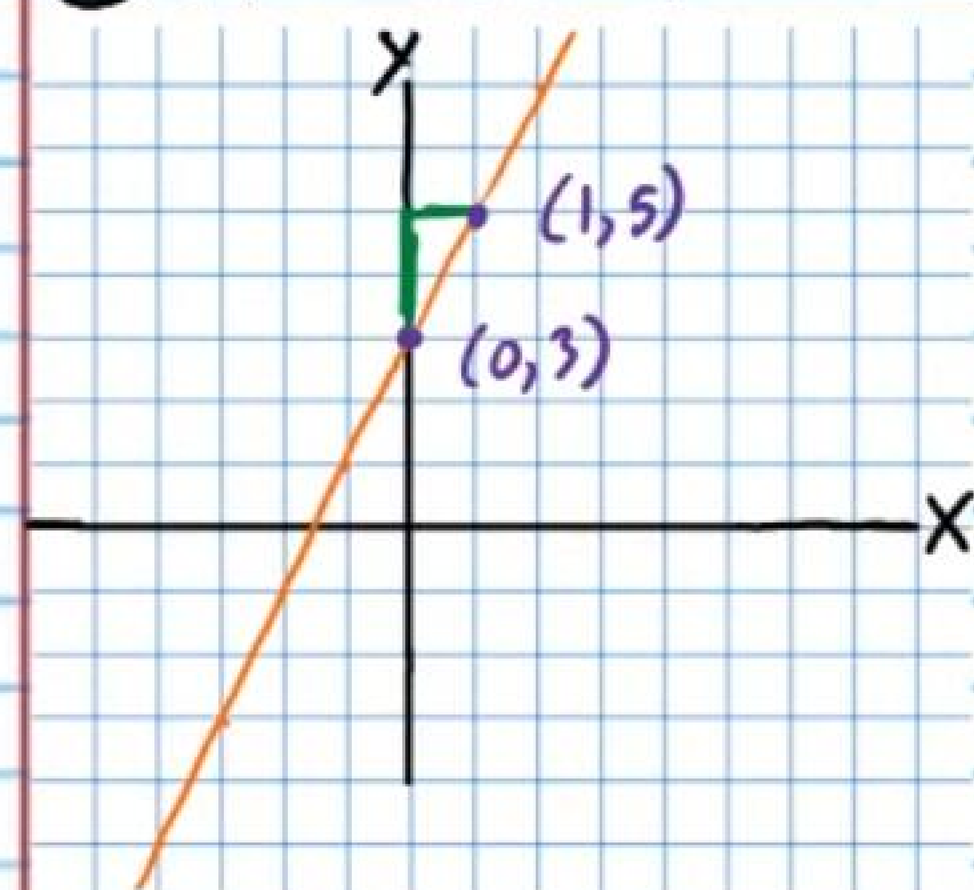
Point I	Point II
Choose $x=0$	Choose $y=0$
$-\frac{0}{3}-2=y$	$-\frac{x}{3}-2=0$
$0-2=y$	$-\frac{x}{3}=2$
$-2=y$	$x=-6$
$(0,-2)$	$(-6,0)$

(40) Method II: $3x+y=3$



Point I	Point II
Choose $x=0$	Choose $y=0$
$3(0)+y=3$	$3x+0=3$
$0+y=3$	$3x=3$
$y=3$	$x=1$
$(0,3)$	$(1,0)$

(41) Method III: $y=2x+3$

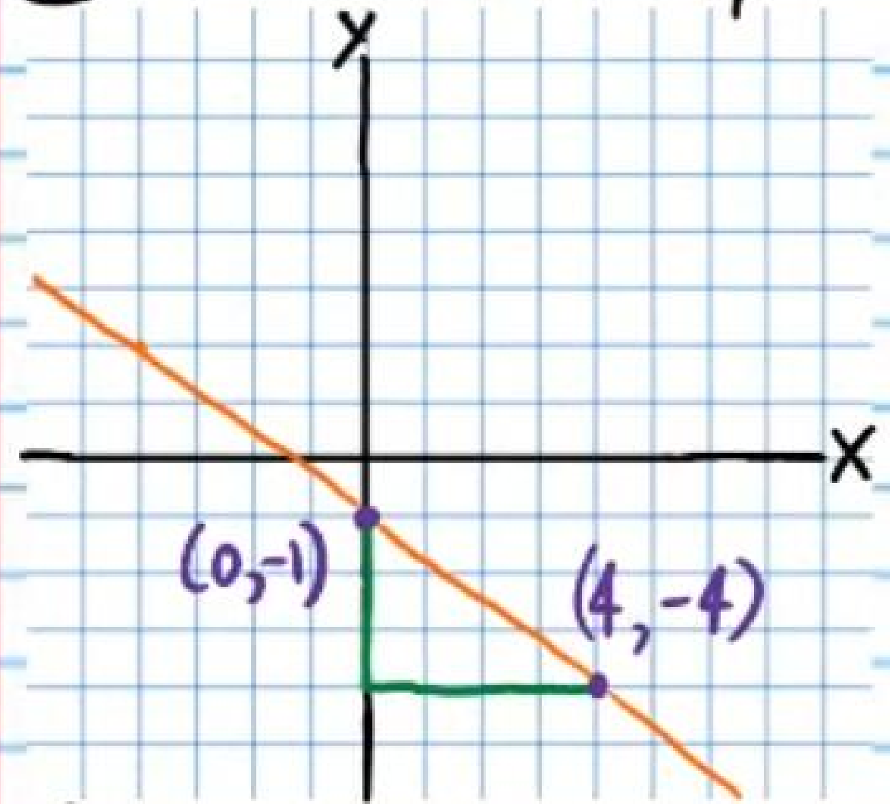


$y=mx+b$

$m=2$, $b=3$

$m=\frac{2}{1}$

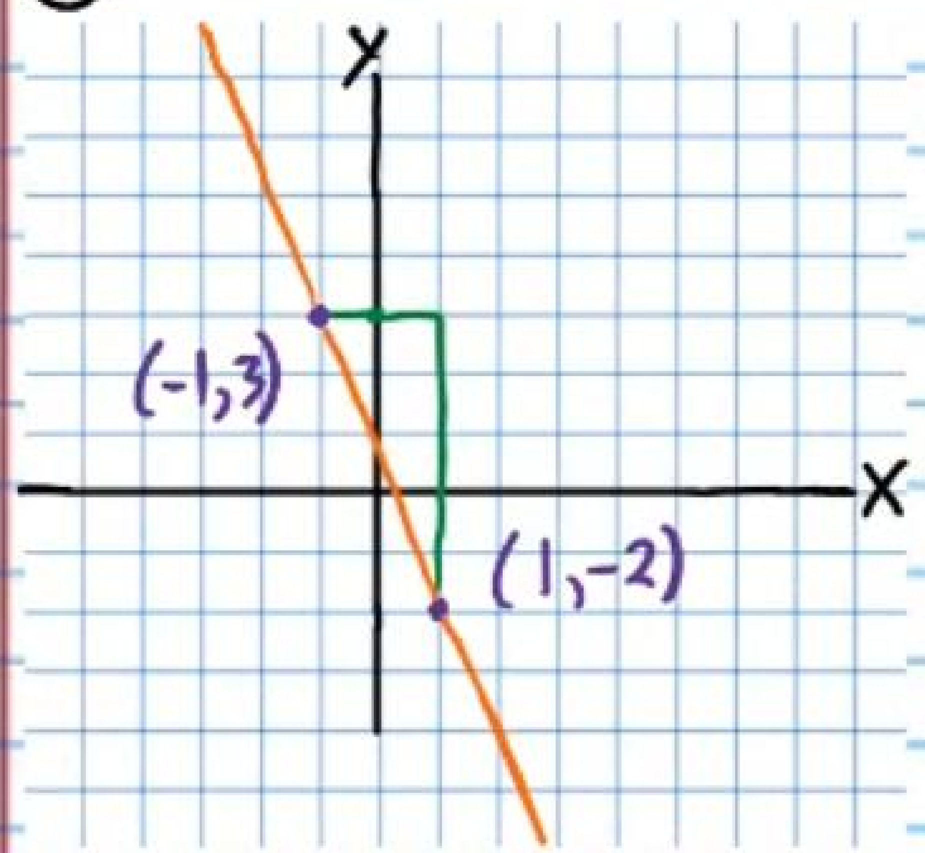
42 Method III: $-\frac{3}{4}x-1=y$



$y=mx+b$

$m=-\frac{3}{4}$, $b=-1$

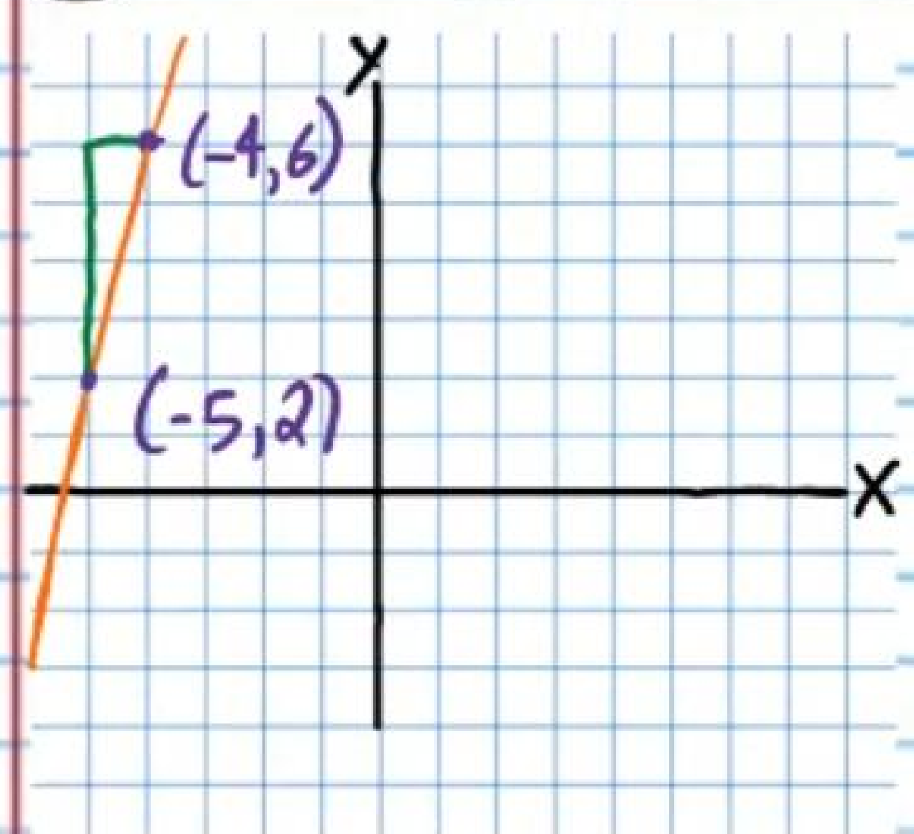
43 Method IV: $y+2=-\frac{5}{2}(x-1)$



$y-y_p=m(x-x_p)$

$m=-5/2$, $(1,-2)$

44 Method IV: $y-2=4(x+5)$



$y-y_p=m(x-x_p)$

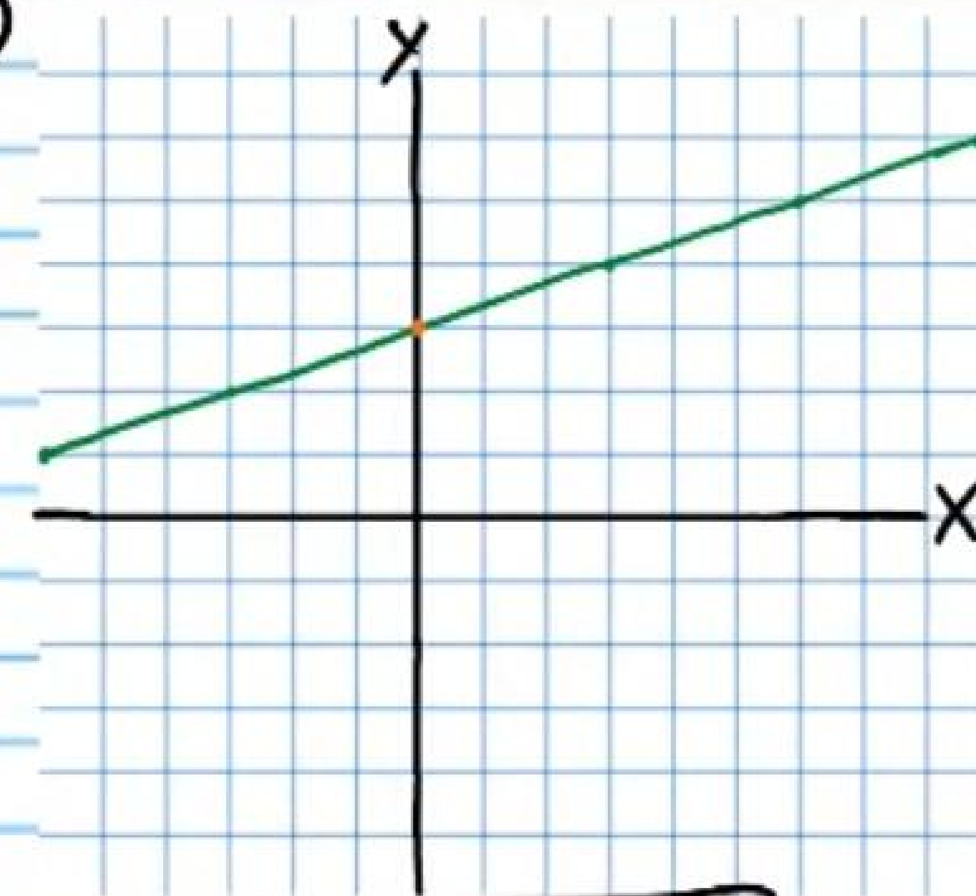
$m=\frac{4}{1}$, $(-5,2)$

Writing Equations of Graphs

Write equations of the lines graphed below.

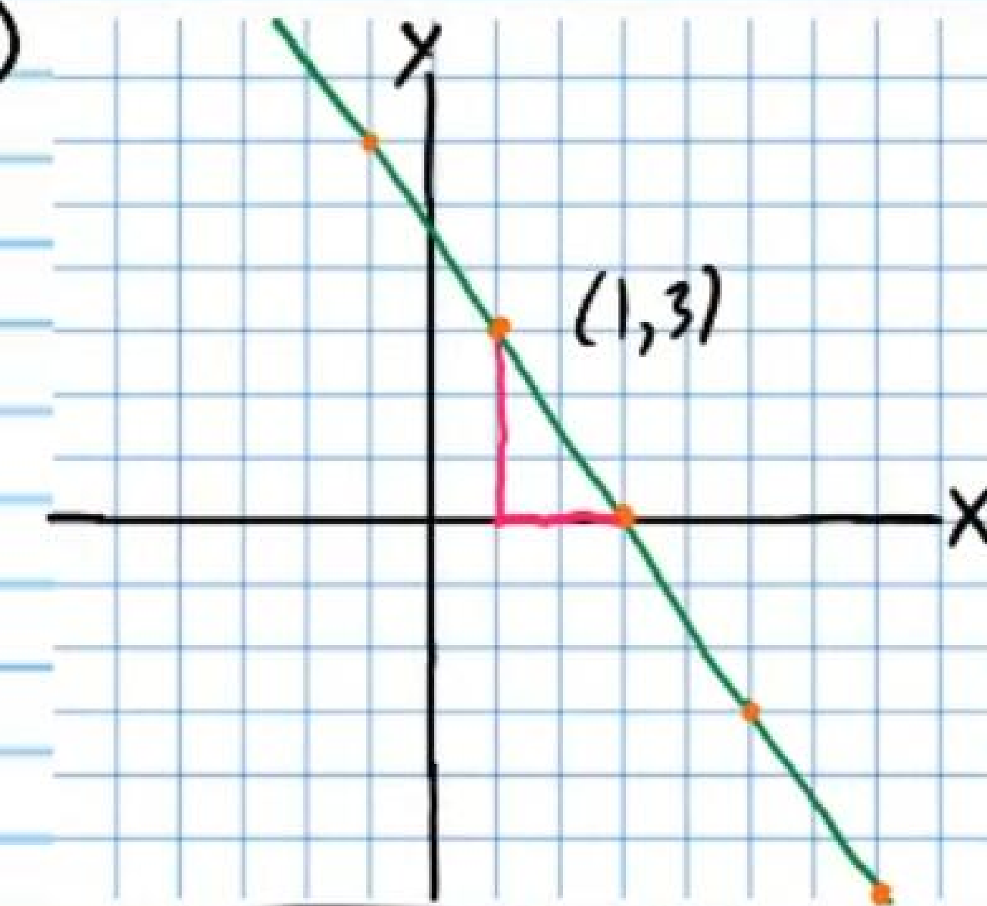
$y=mx+b$

45



$\boxed{y=\frac{1}{3}x+3}$

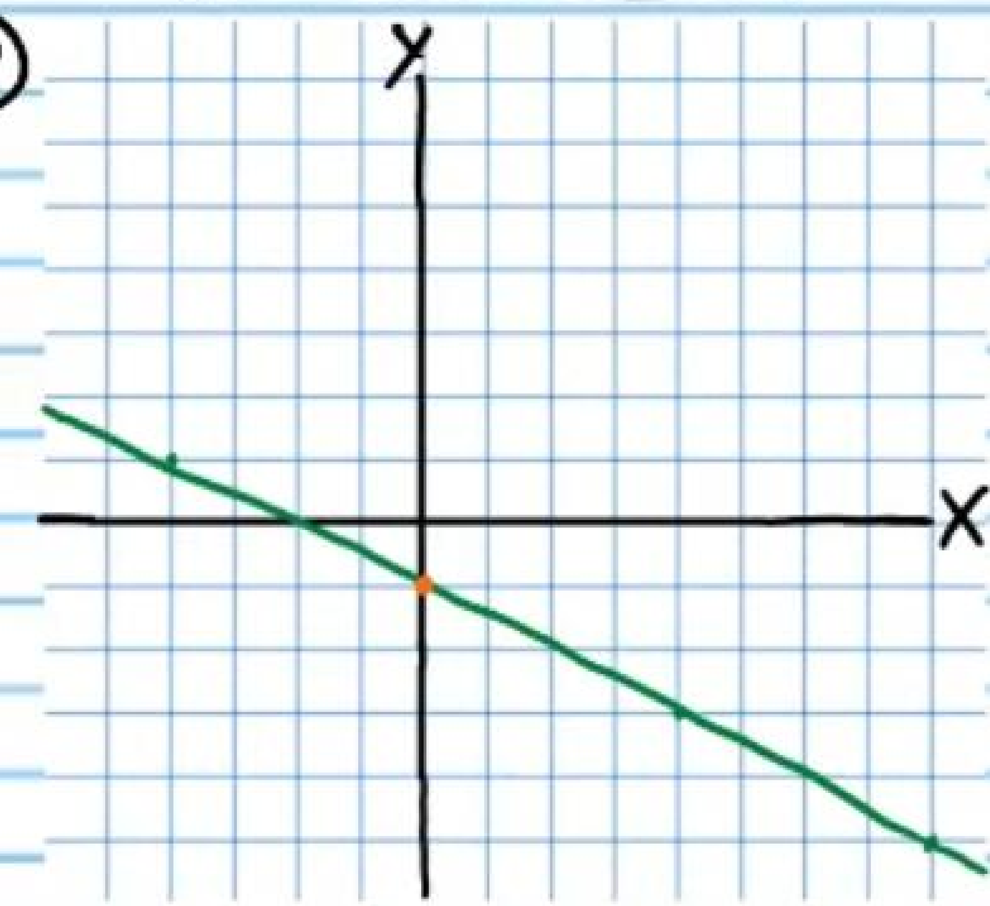
46



$\boxed{y-3=-\frac{3}{2}(x-1)}$

$y-y_p=m(x-x_p)$

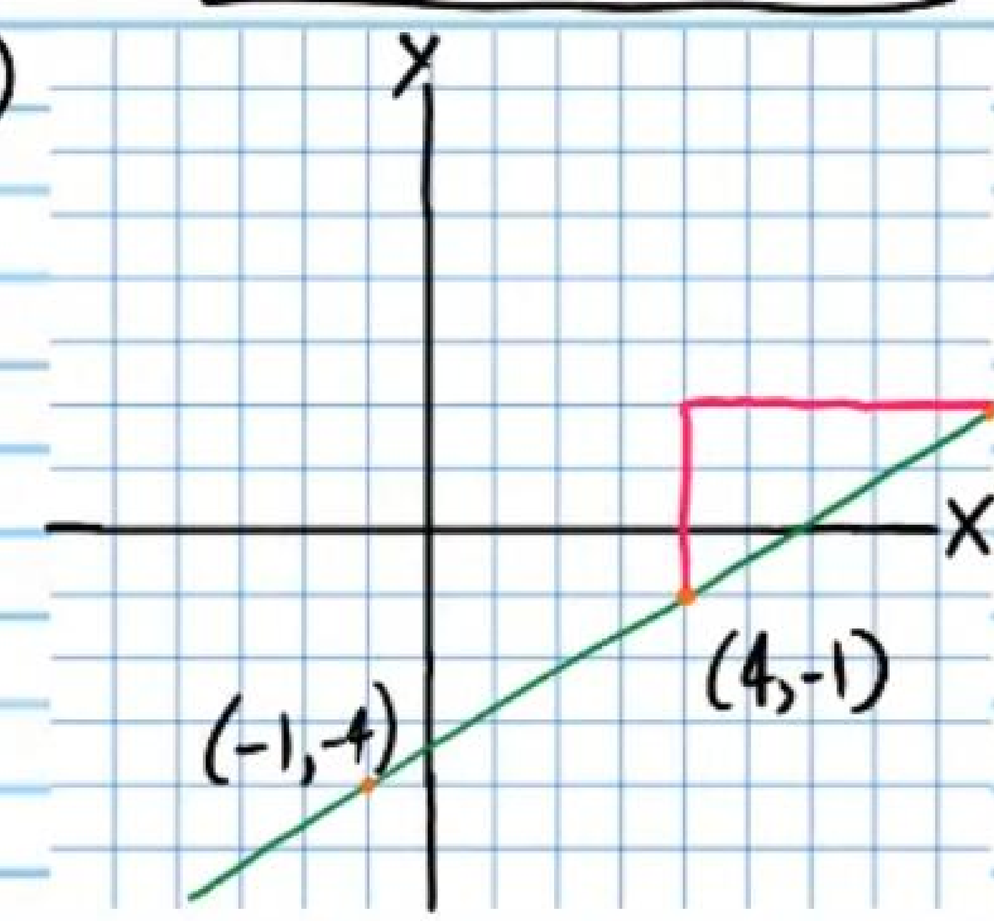
47



$y=-\frac{2}{4}x-1$

$\boxed{y=-\frac{1}{2}x-1}$

48



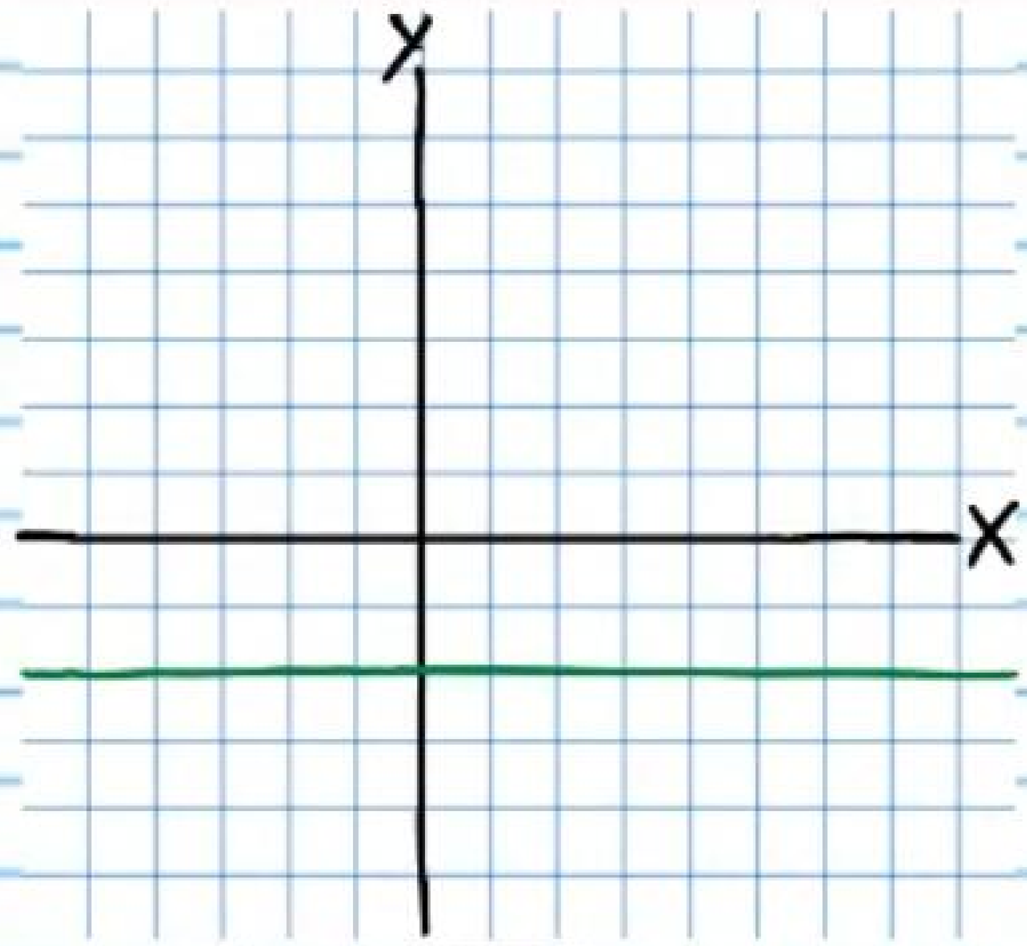
$\boxed{y+1=\frac{3}{5}(x-4)}$

or

$\boxed{y+4=\frac{3}{5}(x+1)}$

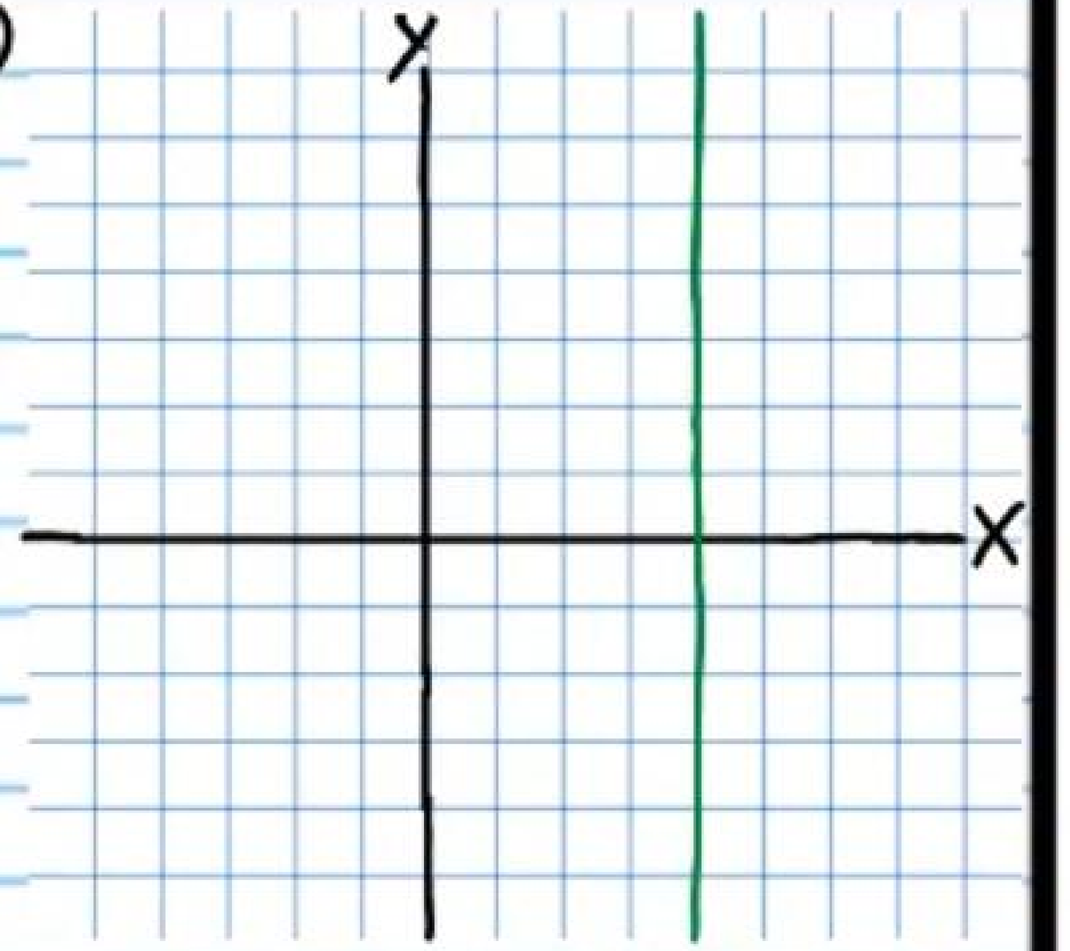
Write equations of the lines graphed below and identify their slopes.

49



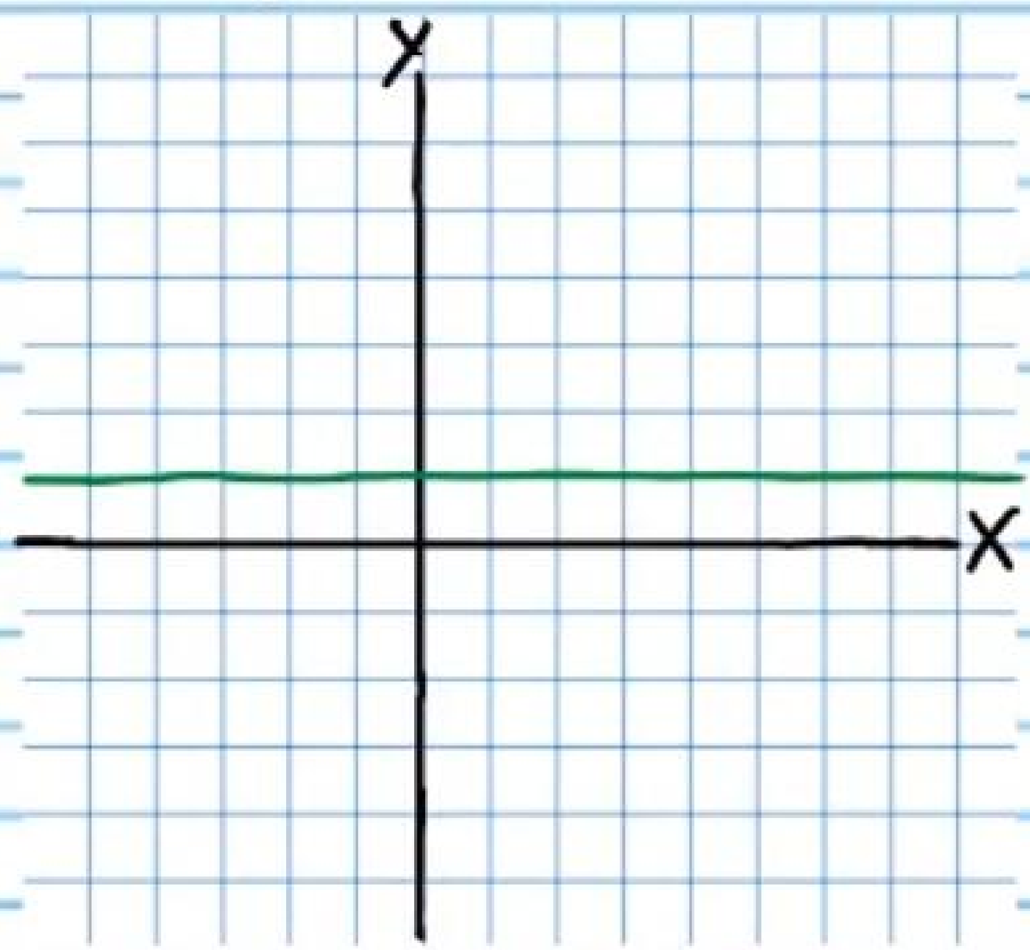
$\boxed{y=-2}$

50



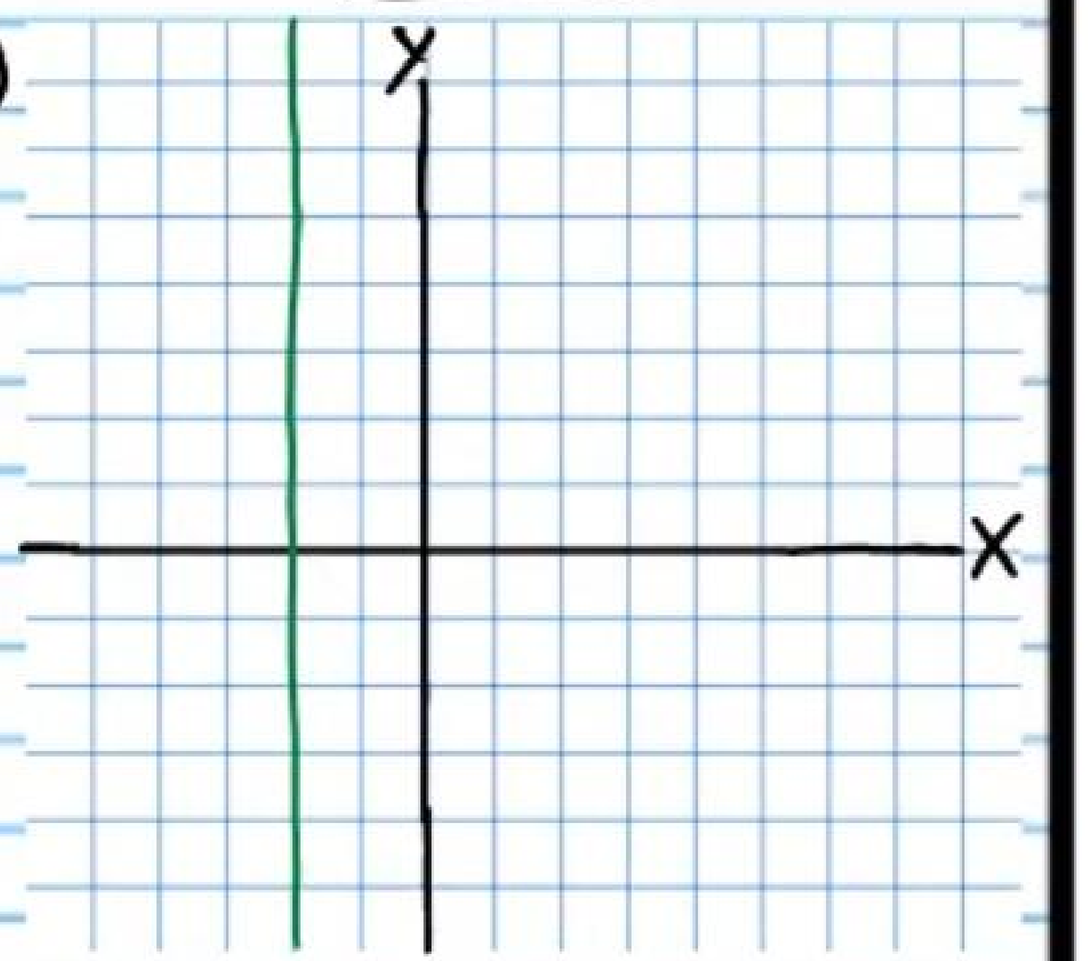
$\boxed{x=4}$

51



$\boxed{y=1}$

52

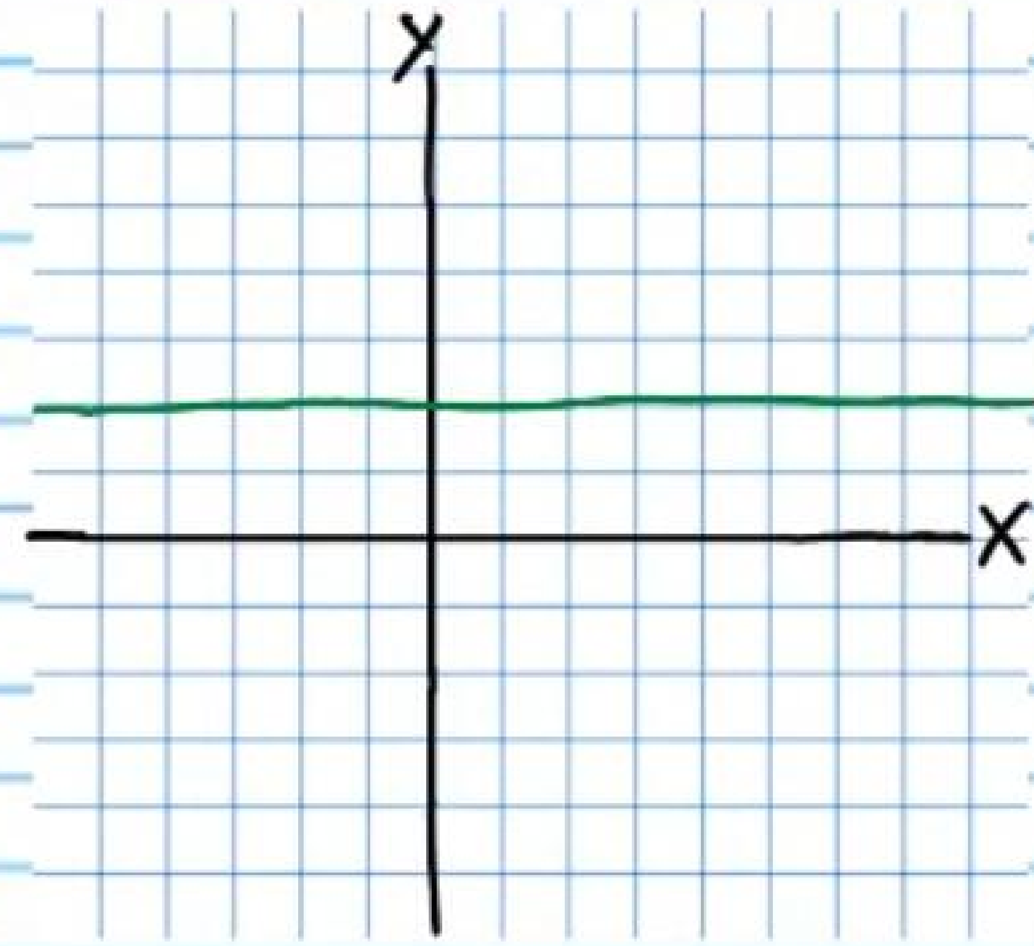


$\boxed{x=-2}$

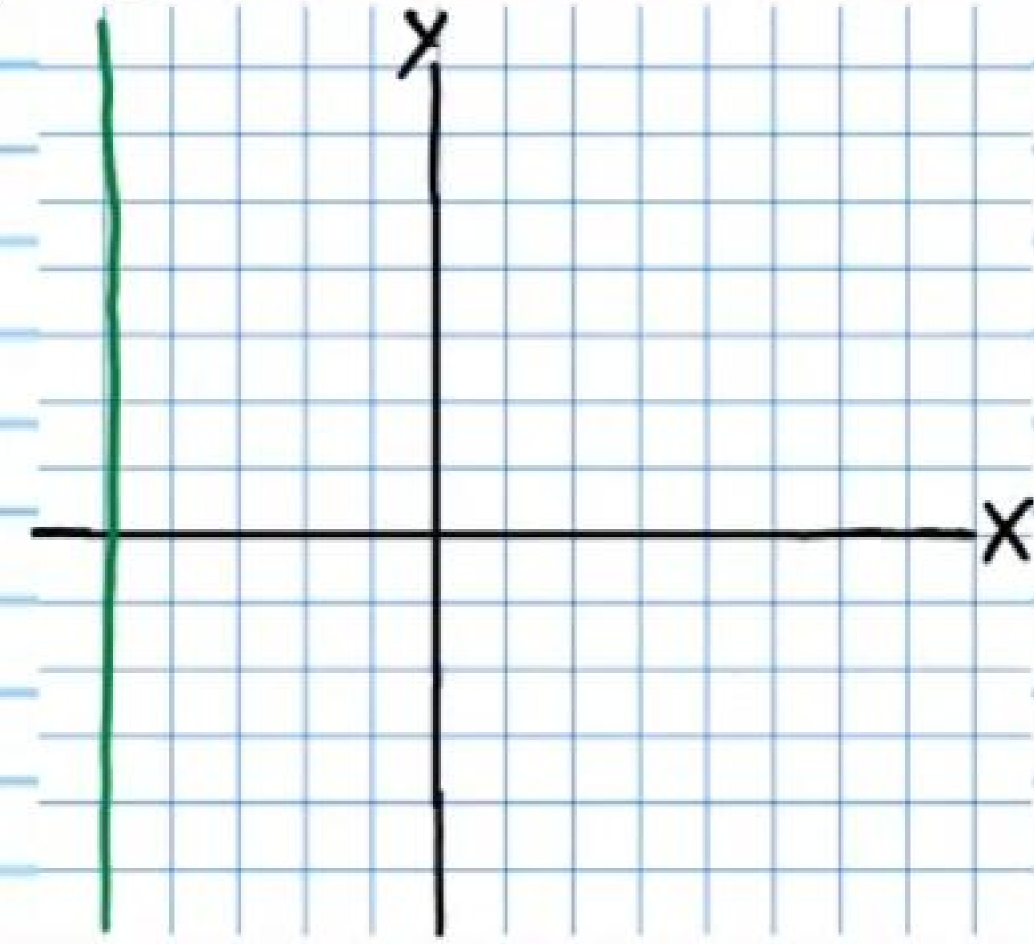
Graphing Vertical or Horizontal Lines

Graph the following equations and state the slopes of the resulting lines.

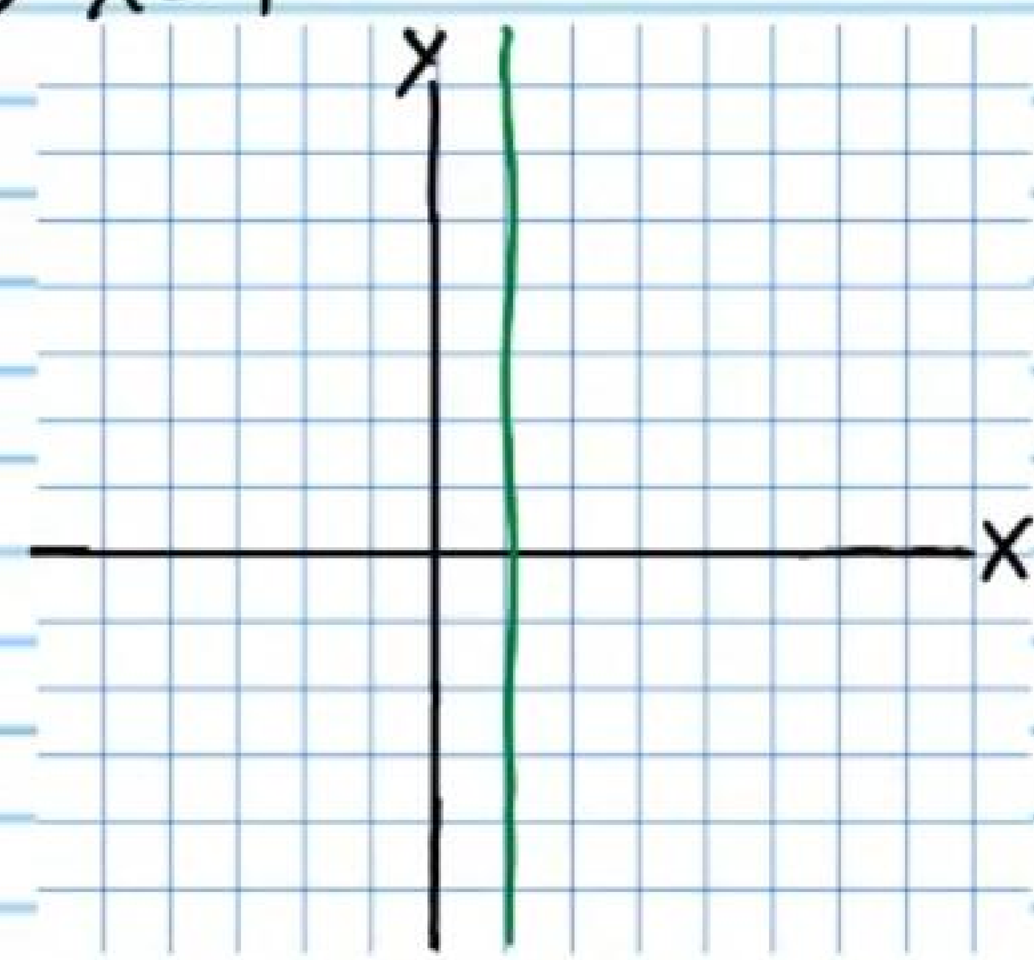
53 $y=2$



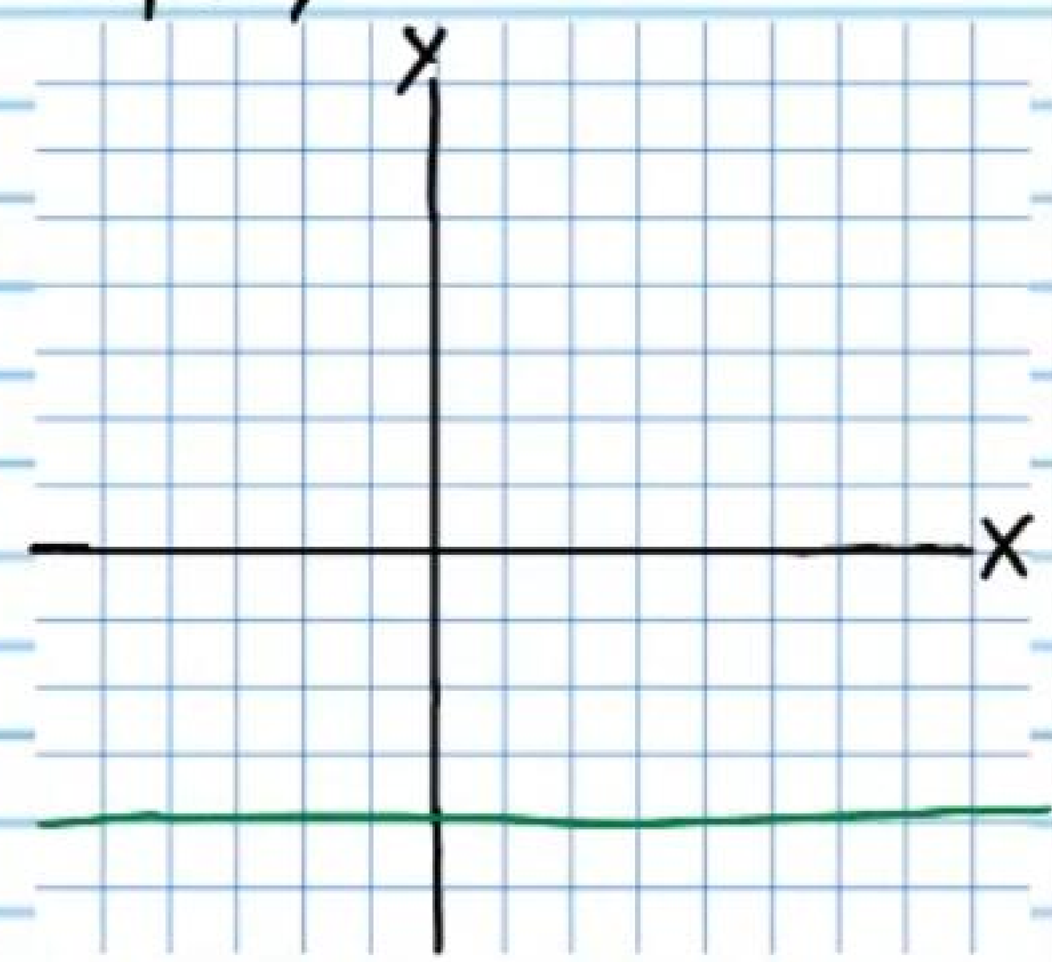
54 $-5=x$



55 $x=1$



56 $-4=y$



Review on Systems of Linear Equations

Solve the following systems of equations with the substitution method.

(57) $\begin{cases} x = 2y \\ x - 3y = 1 \end{cases}$

$$x - 3y = 1$$
$$2y - 3y = 1$$
$$-y = 1$$
$$y = -1$$
$$x = 2(-1)$$
$$x = -2$$
$$\boxed{(-2,-1)}$$

(58) $\begin{cases} -5x + 2y = 2 \\ x = 3y - 4 \end{cases}$

$$-5x + 2y = 2$$
$$-5(3y-4) + 2y = 2$$
$$-15y + 20 + 2y = 2$$
$$-13y + 20 = 2$$
$$y = \frac{18}{13}$$
$$x = 3\left(\frac{18}{13}\right) - 4 = \frac{54}{13} - \frac{4}{1} = 2/13$$
$$\boxed{\left(2/13, \frac{18}{13}\right)}$$

(59) $\begin{cases} 4x + y = 2 \\ x + 1 = y \end{cases}$

$$4x + y = 2$$
$$4x + x + 1 = 2$$
$$5x + 1 = 2$$
$$x = \frac{1}{5}$$
$$\frac{1}{5} + 1 = y$$
$$1\tfrac{1}{5} = y$$
$$\boxed{(1/5, 1\tfrac{1}{5})}$$

(60) $\begin{cases} y = 2x - 3 \\ 4x + 4y = 7 \end{cases}$

$$4x + 4y = 7$$
$$4x + 4(2x-3) = 7$$
$$4x + 8x - 12 = 7$$
$$12x - 12 = 7$$
$$x = \frac{19}{12}$$
$$y = 2\left(\frac{19}{12}\right) - 3$$
$$= 1/6$$
$$\boxed{\left(19/12, \frac{1}{6}\right)}$$

(61) $\begin{cases} 2y = x + 3 \\ 4y = x \end{cases}$

$$2y = x + 3$$
$$2y = 4y + 3$$
$$-2y = 3$$
$$y = -3/2$$
$$4\left(-\frac{3}{2}\right) = x$$
$$-6 = x$$
$$\boxed{\left(-6, -\frac{3}{2}\right)}$$

(62) $\begin{cases} 4 + x = 2y + 1 \\ y = -3x \end{cases}$

$$4 + x = 2y + 1$$
$$4 + x = 2(-3x) + 1$$
$$4 + x = -6x + 1$$
$$4 + 7x = 1$$
$$x = \boxed{-\frac{3}{7}}$$
$$y = -3\left(-\frac{3}{7}\right) = \boxed{\frac{9}{7}}$$

For problem #'s 59-62 you must solve for a variable in one of the equations (in this case y is the easier of the two) before using the substitution method.

(63) $\begin{cases} 3(x-4) = y - 3 \\ y - 7 = 2(x-2) \end{cases}$

$$\begin{cases} 3(x-4) + 3 = y \\ y - 7 = 2(x-2) \end{cases}$$

$$y - 7 = 2(x-2)$$
$$3(x-4) + 3 - 7 = 2(x-2)$$
$$3x - 12 + 3 - 7 = 2x - 4$$
$$3x - 16 = 2x - 4$$
$$x - 16 = -4$$
$$x = 12$$
$$3(x-4) + 3 = y$$
$$3(12-4) + 3 = y$$
$$3(8) + 3 = y$$
$$24 + 3 = y$$
$$27 = y$$
$$\boxed{(12, 27)}$$

(64) $\begin{cases} 2(x-1) = y + 1 \\ -3(x+5) = y - 2 \end{cases}$

$$\begin{cases} 2(x-1) - 1 = y \\ -3(x+5) = y - 2 \end{cases}$$

$$-3(x+5) = y - 2$$
$$-3(x+5) = 2(x-1) - 1 - 2$$
$$-3x - 15 = 2x - 2 - 1 - 2$$
$$-3x - 15 = 2x - 5$$
$$-10 = 5x$$
$$-2 = x$$
$$2(x-1) - 1 = y$$
$$2(-2-1) - 1 = y$$
$$2(-3) - 1 = y$$
$$-7 = y$$
$$\boxed{(-2, -7)}$$

(65) $\begin{cases} y-5=-2(x-3) \\ 6(x+3)=y-3 \end{cases}$

$\begin{cases} y=-2(x-3)+5 \\ 6(x+3)=y-3 \end{cases}$

$6(x+3)=y-3$

$6(x+3)=-2(x-3)+5-3$

$6x+18=-2x+6+5-3$

$6x+18=-2x+8$

$8x=-10$

$x=-\frac{5}{4}$

$y=-2(-\frac{5}{4}-3)+5$

$y=-2\left(\frac{-17}{4}\right)+5$

$y=\frac{17}{2}+5$

$y=\frac{27}{2}$

$\boxed{\left(-\frac{5}{4},\frac{27}{2}\right)}$

(66) $\begin{cases} y+1=2(x+1) \\ y-3=-5(x-2) \end{cases}$

$\begin{cases} y=2(x+1)-1 \\ y-3=-5(x-2) \end{cases}$

$y-3=-5(x-2)$

$2(x+1)-1-3=-5(x-2)$

$2x+2-1-3=-5x+10$

$2x-2=-5x+10$

$7x=12$

$x=\frac{12}{7}$

$y=2(\frac{12}{7}+1)-1$

$y=2\left(\frac{19}{7}\right)-1$

$y=\frac{38}{7}-1$

$y=\frac{31}{7}$

$\boxed{\left(\frac{12}{7},\frac{31}{7}\right)}$